

Interacting with Common Services

Vulnerabilities are commonly discovered by people who use and understand technology, a protocol, or a service. As we move forward, we will find different services to interact with, and we will need to evolve and learn new technology constantly.

To be successful at attacking a service, we need to know its purpose, how to interact with it, what tools we can use, and how to use them. This section will focus on common services and how we can interact with them.

File Share Services

A file sharing service is a type of service that provides, mediates, and monitors the transfer of computer files. Traditionally, we commonly used only internal services for file sharing, such as SMB, NFS, FTP, TFTP, SFTP, but as cloud storage became more popular, we now also have third-party cloud services such as Dropbox, Google Drive, OneDrive, SharePoint, or other file sharing services like Amazon S3, Azure Blob Storage, or Google Cloud Storage. We will be exposed to a mixture of internal and external services, so we need to be familiar with them.

This section will focus on internal services, but this may apply to cloud storage synced locally to servers and workstations.

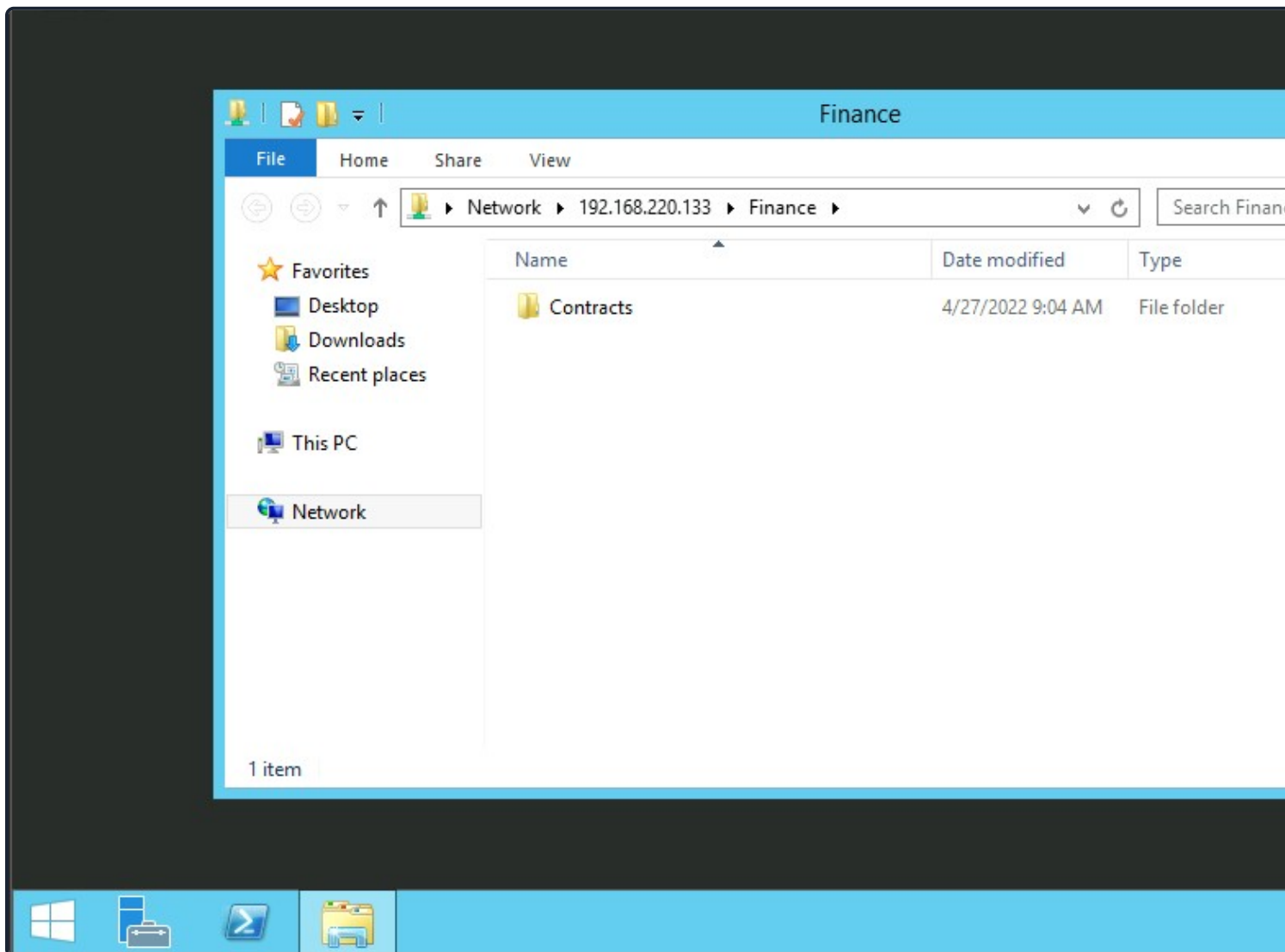
Server Message Block (SMB)

SMB is commonly used in Windows networks, and we will often find share folders in a Windows network. We will explore the GUI, CLI, or tools. Let us cover some common ways of interacting with SMB using Windows & Linux.

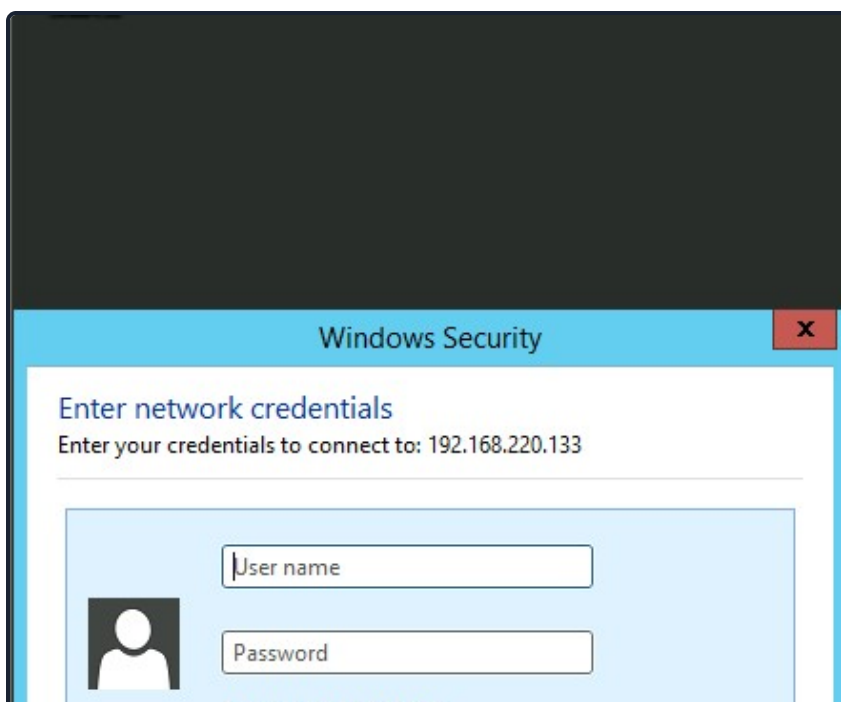
Windows

There are different ways we can interact with a shared folder using Windows, and we will explore a couple of them. We can press `[WINKEY] + [R]` to open the Run dialog box and type the file share location, e.g.: `\\192.168.22.100\share`.

folder. In that case, we will not receive any form of authentication request, and it will display the content of



If we do not have access, we will receive an authentication request.



```
C:\htb> dir \\192.168.220.129\Finance\  
  
Volume in drive \\192.168.220.129\Finance has no label.  
Volume Serial Number is ABCD-EFAA  
  
Directory of \\192.168.220.129\Finance  
  
02/23/2022  11:35 AM    <DIR>          Contracts  
            0 File(s)              4,096 bytes  
            1 Dir(s)  15,207,469,056 bytes free
```

The command `net use` connects a computer to or disconnects a computer from a shared resource or displays computer connections. We can connect to a file share with the following command and map its content to a drive letter.

Windows CMD - Net Use

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```
C:\htb> net use n: \\192.168.220.129\Finance  
  
The command completed successfully.
```

We can also provide a username and password to authenticate to the share.

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```
C:\htb> net use n: \\192.168.220.129\Finance /user:plaintext Password123  
  
The command completed successfully.
```

With the shared folder mapped as the `n:` drive, we can execute Windows commands as if this shared folder was a local drive. We can use the `dir` command to find how many files the shared folder and its subdirectories contain.

Windows CMD - DIR

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```
C:\htb> dir n: /a-d /s /b | find /c ":\\"
```

`/b`

Uses bare format (no heading information or summary)

The following command `| find /c ":\" process the output of dir n: /a-d /s /b to count how many subdirectories. You can use dir /? to see the full help. Searching through 29,302 files is time consuming utilities can help us speed up the search. With dir we can search for specific names in files such as:`

- cred
- password
- users
- secrets
- key
- Common File Extensions for source code such as: .cs, .c, .go, .java, .php, .as

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```
C:\htb>dir n:\*cred* /s /b  
  
n:\Contracts\private\credentials.txt  
  
C:\htb>dir n:\*secret* /s /b  
  
n:\Contracts\private\secret.txt
```

If we want to search for a specific word within a text file, we can use [findstr](#).

Windows CMD - Findstr

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```
c:\htb>findstr /s /i cred n:\*.*  
  
n:\Contracts\private\secret.txt:file with all credentials  
n:\Contracts\private\credentials.txt:admin:SecureCredentials!
```

We can find more [findstr](#) examples [here](#).

Windows PowerShell

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```
PS C:\htb> New-PSDrive -Name "N" -Root "\\192.168.220.129\Finance" -PSProvider "FileSystem"
```

Name	Used (GB)	Free (GB)	Provider	Root
----	-----	-----	-----	----
N			FileSystem	\\192.168.220.129\Finance

To provide a username and password with Powershell, we need to create a [PSCredential object](#). It offers a way to store usernames, passwords, and credentials.

Windows PowerShell - PSCredential Object

Interacting with Common Services

```
PS C:\htb> $username = 'plaintext'
PS C:\htb> $password = 'Password123'
PS C:\htb> $secpassword = ConvertTo-SecureString $password -AsPlainText -Force
PS C:\htb> $cred = New-Object System.Management.Automation.PSCredential $username, $secpassword
PS C:\htb> New-PSDrive -Name "N" -Root "\\192.168.220.129\Finance" -PSProvider "FileSystem"
```

Name	Used (GB)	Free (GB)	Provider	Root
----	-----	-----	-----	----
N			FileSystem	\\192.168.220.129\Finance

In PowerShell, we can use the command [Get-ChildItem](#) or the short variant [gci](#) instead of the command [ls](#).

Windows PowerShell - GCI

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```
PS C:\htb> N:
PS N:\> (Get-ChildItem -File -Recurse | Measure-Object).Count

29302
```

We can use the property [-Include](#) to find specific items from the directory specified by the Path parameter.

```
N:\Contracts\private\credentials.txt:1:admin:SecureCredentials!
```

CLI enables IT operations to automate routine tasks like user account management, nightly backups, or in general, you can perform operations more efficiently by using scripts than the user interface or GUI.

Linux

Linux (UNIX) machines can also be used to browse and mount SMB shares. Note that this can be done with a Linux machine or a Samba server. Even though some Linux distributions support a GUI, we will focus on the command line and tools to interact with SMB. Let's cover how to mount SMB shares to interact with directories and files.

Linux - Mount

Interacting with Common Services

```
dbcyph0n@htb[/htb]$ sudo mkdir /mnt/Finance
dbcyph0n@htb[/htb]$ sudo mount -t cifs -o username=plaintext,password=Password123,domain=.
```

As an alternative, we can use a credential file.

Interacting with Common Services

```
dbcyph0n@htb[/htb]$ mount -t cifs //192.168.220.129/Finance /mnt/Finance -o credential=/mnt/credentials.txt
```

The file `credentialfile` has to be structured like this:

CredentialFile

Code: `txt`

```
username=plaintext
password=Password123
domain=.
```

Note: We need to install `cifs-utils` to connect to an SMB share folder. To install it we can execute from terminal:

```
install cifs-utils
```

```
/mnt/Finance/Contracts/private/credentials.txt:1:admin:SecureCredentials!  
/mnt/Finance/Contracts/private/secret.txt:1:file with all credentials
```

Other Services

There are other file-sharing services such as FTP, TFTP, and NFS that we can attach (mount) using different protocols. However, once we mount a file-sharing service, we must understand that we can use the available tools in Linux to interact with files and directories. As we discover new file-sharing services, we will need to investigate how they work and how to interact with them.

Email

We typically need two protocols to send and receive messages, one for sending and another for receiving. The Simple Mail Transfer Protocol (SMTP) is an email delivery protocol used to send mail over the internet. Likewise, a supporting protocol is needed to retrieve an email from a service. There are two main protocols we can use POP3 and IMAP.

We can use a mail client such as [Evolution](#), the official personal information manager, and mail client for the GNOME desktop environment. We can interact with an email server to send or receive messages with a mail client. To install Evolution, use the following command:

Linux - Install Evolution

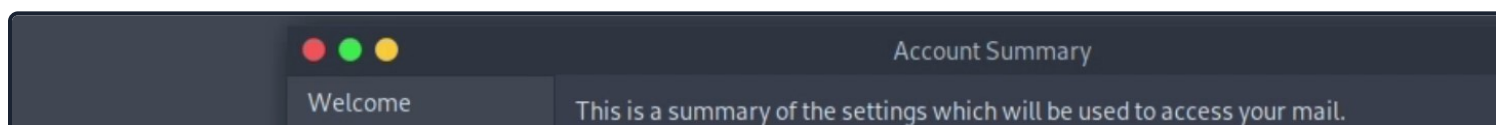
Interacting with Common Services

```
dbcypth0n@htb[/htb]$ sudo apt-get install evolution  
...SNIP...
```

Note: If an error appears when starting evolution indicating "bwrap: Can't create file at ...", use this command:
`WEBKIT_FORCE_SANDBOX=0 && evolution.`

Video - Connecting to IMAP and SMTP using Evolution

Click on the image below to see a short video demonstration.



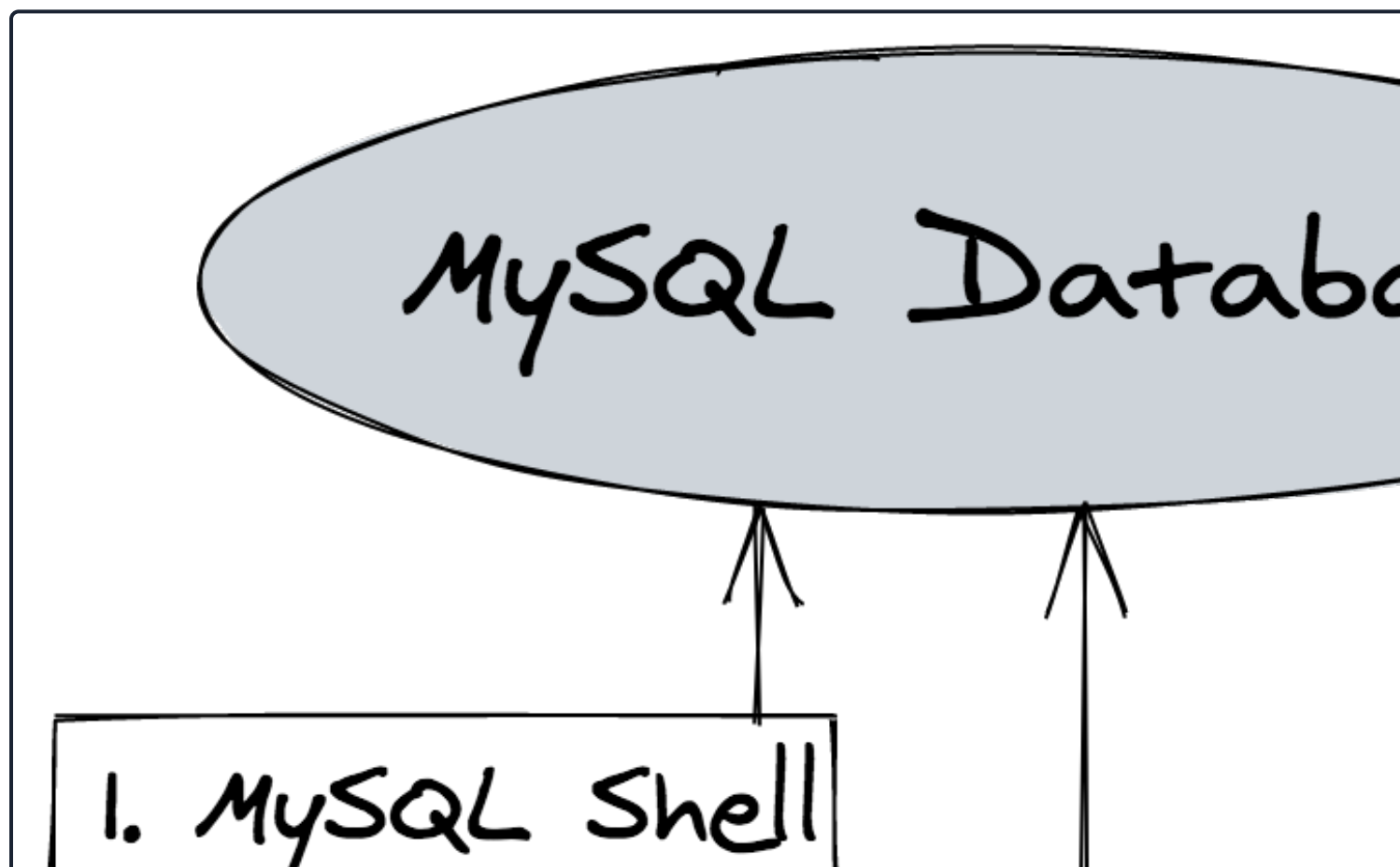
We can use the domain name or IP address of the mail server. If the server uses SMTPS or IMAPS, we'll need the STARTTLS method (TLS on a dedicated port or STARTTLS after connecting). We can use the [Check for Supported Authentication Methods](#) method to confirm if the server supports our selected method.

Databases

Databases are typically used in enterprises, and most companies use them to store and manage information. There are many types of databases, such as Hierarchical databases, NoSQL (or non-relational) databases, and SQL relational databases. The two most common relational databases are MySQL & MSSQL. We have three ways to interact with databases:

1. Command Line Utilities ([mysql](#) or [sqsh](#))
2. A GUI application to interact with databases such as HeidiSQL, MySQL Workbench, or SQL Server Management Studio
3. Programming Languages

MySQL example



Command Line Utilities

MSSQL

To interact with [MSSQL \(Microsoft SQL Server\)](#) with Linux we can use [sqsh](#) or [sqlcmd](#) if you are using Windows. [sqlcmd](#) is a friendly prompt. It is intended to provide much of the functionality provided by a command shell, such as pipes, back-grounding, job control, history, command substitution, and dynamic configuration. We can start an interactive session as follows:

Linux - SQSH

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```
dbcyph0n@htb[/htb]$ sqsh -S 10.129.20.13 -U username -P Password123
```

The [sqlcmd](#) utility lets you enter Transact-SQL statements, system procedures, and script files through a variety of methods:

- At the command prompt.
- In Query Editor in SQLCMD mode.
- In a Windows script file.
- In an operating system (Cmd.exe) job step of a SQL Server Agent job.

Windows - SQLCMD

Interacting with Common Services

```
C:\htb> sqlcmd -S 10.129.20.13 -U username -P Password123
```

To learn more about [sqlcmd](#) usage, you can see [Microsoft documentation](#).

MySQL

To interact with [MySQL](#), we can use MySQL binaries for Linux ([mysql](#)) or Windows ([mysql.exe](#)). MySQL comes pre-installed on most Linux distributions, but we can install MySQL binaries for Linux or Windows using this [guide](#). Start an interactive session as follows:

Linux - MySQL

supports connecting to multiple database engines such as MSSQL, MySQL, PostgreSQL, among others, attacker, to interact with common database servers.

To install [dbeaver](#) using a Debian package we can download the release .deb package from <https://github.com> and execute the following command:

Install dbeaver

Interacting with Common Services

```
dbcypthon@htb[/htb]$ sudo dpkg -i dbeaver-<version>.deb
```

To start the application use:

Run dbeaver

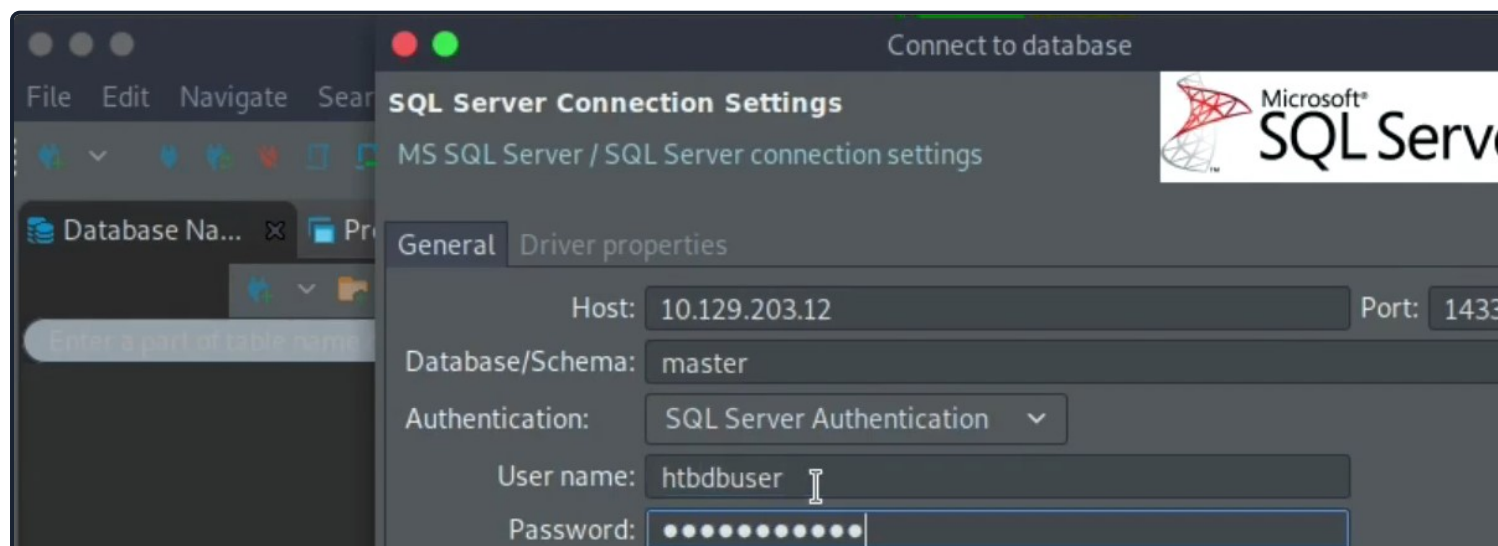
Interacting with Common Services

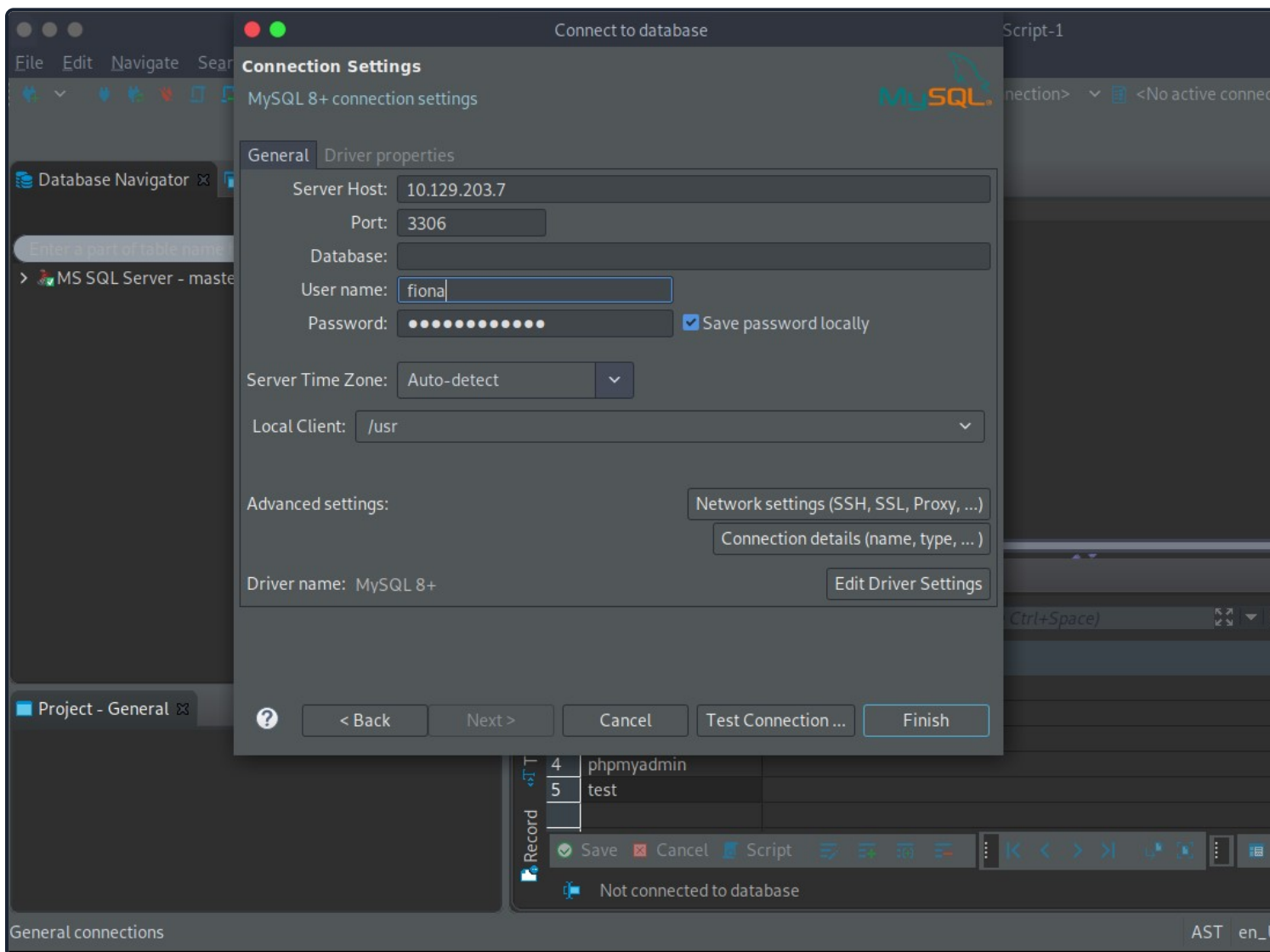
```
dbcypthon@htb[/htb]$ dbeaver &
```

To connect to a database, we will need a set of credentials, the target IP and port number of the database are trying to connect to (MySQL, MSSQL, or another).

Video - Connecting to MSSQL DB using dbeaver

Click on the image below for a short video demonstration of connecting to an MSSQL database using [dbeaver](#)





Once we have access to the database using a command-line utility or a GUI application, we can use commands to enumerate databases and tables containing sensitive information such as usernames and passwords. If we could potentially execute commands as the MSSQL service account. Later in this module, we will discuss statements and attacks for MSSQL & MySQL databases.

Tools

It is crucial to get familiar with the default command-line utilities available to interact with different services. In the field, we will find tools that can help us be more efficient. The community commonly creates those tools. We will have ideas on how a tool can be improved or for creating our own tools, even if we are not full-time developers. We are familiar with hacking. The more we learn, the more we find ourselves looking for a tool that does not exist. We learn and create our tools.

Tools to Interact with Common Services

General Troubleshooting

Depending on the Windows or Linux version we are working with or targetting, we may encounter different errors when trying to connect to a service.

Some reasons why we may not have access to a resource:

- Authentication
- Privileges
- Network Connection
- Firewall Rules
- Protocol Support

Keep in mind that we may encounter different errors depending on the service we are targetting. We can use the internet to our advantage and search for official documentation or forums where people solved an issue similar to ours.

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 Attacking Common Services - Easy

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 Attacking Common Services - Hard

My Workstation

O F F L I N E

 Start Instance

 / 1 spawns left